

TrES-5b

R-1846

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_210826 · created 2026-08-03T07:39:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459452.731 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.026
Transit depth (Rp/Rs)^2	3.42 +/- 0.97 [%]
Orbital Inclination (inc)	82.92 +/- 2.21
Airmass coefficient 1 (a1)	0.571 +/- 0.019
Airmass coefficient 2 (a2)	0.47 +/- 0.22
Scatter in the residuals of the lightcurve fit is	3.37 %
Best Comparison Star	#3 - [83, 67]
Optimal Aperture	3.0
Optimal Annulus	9.7
Transit Duration (day)	0.064 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.026 vs 0.142 (deviation 1.37σ)
Duration fit vs published	0.064 d vs 0.0758 d (deviation 0.47σ)
Mid-time O–C	1.0 min
Depth SNR	5.9
Residual scatter	3.37 %

Light curve

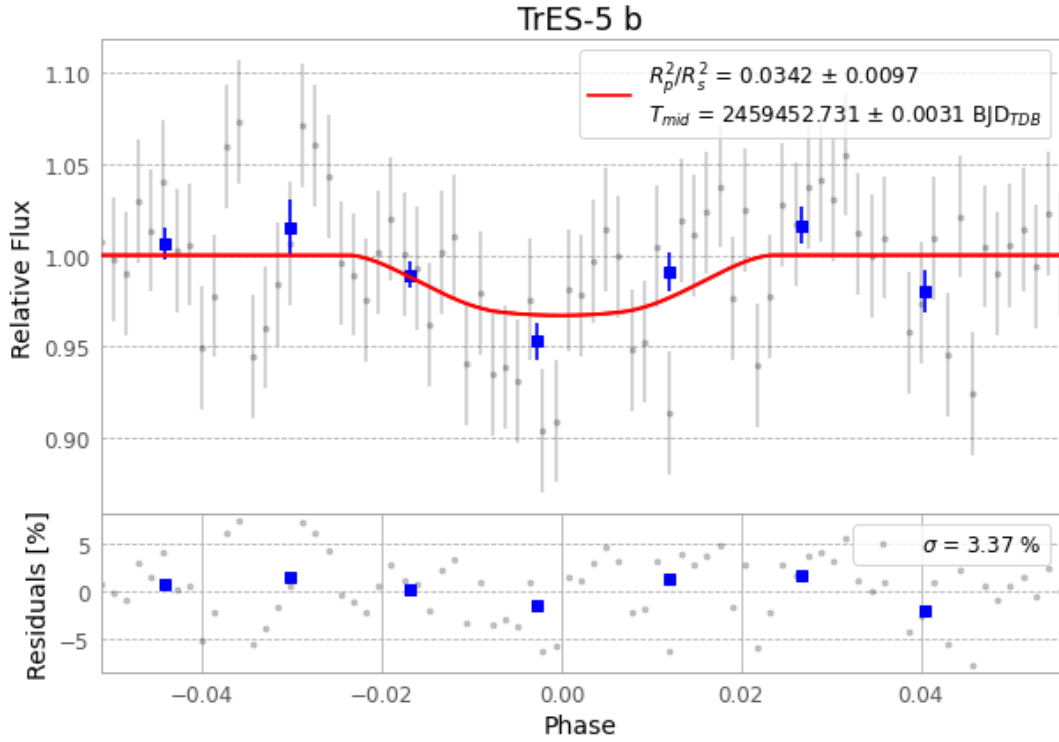


Figure 1 — FinalLightCurve_TrES-5 b_2021-08-25.png (SHA-256 842f8b76578c17aa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 59b6cf114fdbeb55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), TRES-5210826033912.FITS → TRES-5210826072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-210826.log (01b4e3c2e14e...), FinalLightCurve_TrES-5 b_2021-08-25.png (842f8b76578c...), FinalLightCurve_TrES-5 b_2021-08-25.csv (348776a7090b...)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 07:39Z.